Smart City Project Proposal: F-Guard

F-Guard: An Intelligent Solar-Powered Forest Monitoring System for Early Detection of Illegal Deforestation

ABSTRAK

Maraknya deforestasi ilegal di Indonesia memicu krisis lingkungan yang mengancam keanekaragaman hayati, kestabilan iklim, dan kesejahteraan masyarakat. Kelemahan sistem pengawasan konvensional seperti patroli lapangan dan citra satelit, terutama di daerah terpencil, menuntut solusi berbasis teknologi yang adaptif dan berkelanjutan. F-Guard merupakan sistem pemantauan hutan cerdas bertenaga surya yang dirancang untuk mendeteksi aktivitas deforestasi ilegal secara otomatis dan real-time. Sistem ini mengintegrasikan sensor suara, pemrosesan fitur audio (MFCC), dan model deep learning berbasis CNN untuk mengenali suara gergaji mesin dari lingkungan alami. Energi diperoleh dari panel surya, menjadikan perangkat ini mandiri dan ramah lingkungan. Komunikasi data dilakukan melalui jaringan efisien seperti LoRa atau Sigfox, memungkinkan pengiriman lokasi dan notifikasi langsung ke aplikasi mobile milik petugas kehutanan. F-Guard dilengkapi dengan GPS, sensor getar anti-sabotase, dan desain fisik yang menyerupai batang pohon untuk kamuflase. Selain aspek teknis, pengembangan F-Guard juga mencakup model bisnis berbasis keberlanjutan dengan segmen pelanggan meliputi pemerintah daerah, LSM lingkungan, hingga perusahaan kehutanan sebagai bagian dari program CSR/ESG. Inovasi ini tidak hanya meningkatkan efisiensi pengawasan hutan, tetapi juga mendukung pengembangan kota cerdas (smart city) yang berorientasi pada keberlanjutan dan keamanan lingkungan.

Kata kunci: deforestasi ilegal, deep learning, Internet of Things, energi surya, sistem pemantauan hutan

ABSTRACT

The rise of illegal deforestation in Indonesia has triggered an environmental crisis that threatens biodiversity, climate stability, and community welfare. The limitations of conventional monitoring systems such as field patrols and satellite imagery—especially in remote areas—demand adaptive and sustainable technology-based solutions. F-Guard is an intelligent, solar-powered forest monitoring system designed to automatically and in real-time detect illegal deforestation activities. This system integrates sound sensors, audio feature processing (MFCC), and a deep learning model based on CNN to recognize chainsaw sounds in natural environments. Powered by solar panels, the device is energy-independent and environmentally friendly. Data communication is conducted via efficient networks such as LoRa or Sigfox, enabling real-time location tracking and notifications to be sent directly to forest officers through a mobile application. F-Guard is equipped with GPS, an anti-tampering

vibration sensor, and a tree-trunk-like camouflage design to blend into its surroundings. Beyond the technical aspects, F-Guard also includes a sustainability-oriented business model targeting customer segments such as local governments, environmental NGOs, and forestry companies as part of their CSR/ESG programs. This innovation not only enhances the efficiency of forest monitoring but also supports the development of smart cities focused on sustainability and environmental security.

Keywords: illegal deforestation, deep learning, Internet of Things, solar energy, forest monitoring system

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CHAPTER ONE

1.1 Background Problem

Deforestation has become a global environmental crisis, threatening biodiversity, climate stability, and community well-being. According to Global Forest Watch (2023), the world loses forest area equivalent to 11 football fields every minute. In 2023, the global loss of tropical primary forest reached 3.7 million hectares, resulting in 2.4 gigatons of carbon emissions (World Resources Institute, 2024). In Indonesia, tree cover loss surged from 900 thousand hectares in 2022 to 1.4 million hectares in 2023, exacerbating floods, landslides, and droughts in areas surrounding forests.

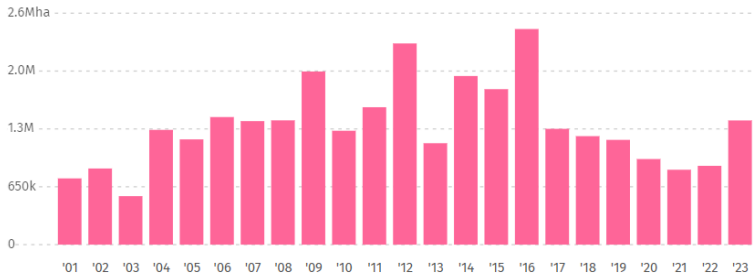


Figure 1. Indonesian Tree Cover Loss Graph

The main factors contributing to deforestation are illegal logging, land conversion for plantations, and weak law enforcement. Illegal logging using chainsaws is the most common practice (Abimanyu, 2022) and is difficult to detect due to limited field patrols, extreme weather, and forest canopy obstruction for satellite imagery. Conventional surveillance systems have proven ineffective in responding to these threats in real time. Therefore, an adaptive and sustainable technology-based approach is needed. Technologies such as deep learning and the Internet of Things (IoT) can be used to automatically detect the sounds of illegal activities, identify chainsaw sounds amidst natural sounds, and send immediate alerts to authorities.

However, the effectiveness of such systems also depends on their energy independence. Remote locations often lack access to electricity, necessitating environmentally friendly and sustainable solutions. This is where renewable energy,

particularly solar panels, plays a crucial role. By utilizing solar power, the system can operate continuously without relying on fossil fuels.

This innovation is embodied in F-Guard, a solar-powered forest monitoring system equipped with sound sensors and deep learning technology, integrated with a mobile application. F-Guard enables automatic detection of illegal activities and instant delivery of information to officers, supporting information mobility and rapid response in environmental conservation efforts. By relying on renewable energy and an environmentally friendly system, F-Guard is not only efficient and sustainable but also introduces an innovative, adaptive approach to forest surveillance, becoming an integral part of data-driven smart city development oriented towards a secure, green future.

1.2 Purpose

a. Develop a solar-powered monitoring tool for illegal deforestation activities, utilizing solar panels as the primary energy source to enable autonomous operation in remote areas.
b. Implement Internet of Things (IoT) and deep learning technologies to automatically and accurately detect illegal logging activities.
c. Provide a mobile application as a monitoring platform that supports information mobility and rapid response from authorities.

1.3 Target Users

F-Guard is designed for institutions and stakeholders directly involved in forest protection and environmental monitoring. The primary users of the mobile application and alert system include:

a. Government officers from Forestry and Environmental Agencies (KLHK, DLHK)
b. Conservation patrol teams and forest rangers
c. Disaster response teams (BPBD)

CHAPTER TWO

2.1 Product Analysis

The issue of illegal deforestation in Indonesia remains a significant challenge that is difficult to tackle with conventional methods such as manual patrols or satellite surveillance. Field patrols require many personnel, face difficult access, and are less effective for widespread monitoring. Meanwhile, satellite technology tends to be expensive and has limitations in penetrating the forest canopy, especially in bad weather conditions.

F-Guard emerges as a solution that addresses these weaknesses through a smart and sustainable technological approach. This tool is designed as an illegal deforestation monitoring system based on sound sensors, deep learning, and solar energy, operating actively, autonomously, and in real-time in remote locations without relying on conventional electricity or public internet connections.

2.1.1 Key Features of F-Guard

a. Sound sensor to capture chainsaw sounds in real-time.
b. FFT pre-processing and MFCC Feature Extraction performed by the ESP32 microcontroller to convert raw audio signals into feature data ready for transmission.
c. ESP32 microcontroller that controls device workflow, performs audio pre-processing, and sends feature data to a server or gateway for classification.
d. Deep learning model (CNN + MFCC) running on a local server or gateway, performing sound classification based on MFCC features.
e. Solar panels and Li-Ion battery providing independent power without the need for a conventional electricity supply.
f. GPS for location tracking when detection occurs.
g. Sigfox/LoRa communication module for energy-efficient data transmission from remote locations.
h. Mobile application as a monitoring hub, displaying notifications, device status, and location maps.
i. Anti-sabotage vibration sensor to prevent theft or unauthorized relocation of the device.

2.1.2 Advantages of F-Guard

a. Energy independent, not reliant on conventional electricity or internet networks.
b. High efficiency, capable of operating 24/7 with low power consumption.
c. Automatic notifications sent when illegal activity is detected.
d. Easy to install in remote locations.
e. Environmentally friendly, using renewable energy and energy-efficient materials.
f. Scalable, allowing for gradual expansion.

2.1.3 Disadvantages and Challenges of F-Guard

a. Higher initial development and production costs compared to conventional systems.
b. Dependence on communication signals (Sigfox/LoRa) which are not yet evenly distributed across all regions.
c. Sensor performance can be disrupted by extreme environmental noise such as heavy rain or strong winds.
d. Requires periodic maintenance for solar panels.

CHAPTER THREE

3.1 Business Model

To ensure the sustainability and scalability of F-Guard, a business model has been developed based on the Business Model Canvas (BMC) framework, which includes nine main elements:

a. *Customer Segments*

Local governments (especially conservation areas), Forestry and Environmental Agencies, environmental NGOs (e.g., WWF, WALHI), indigenous communities or forest eco-tourism villages, and palm oil or forestry companies (as part of CSR/ESG initiatives).

b. *Value Propositions*

An intelligent, energy-independent, and environmentally friendly forest monitoring solution. F-Guard can detect illegal deforestation activities automatically and in real-time using sound sensors and artificial intelligence. Powered by solar panels, the device can operate without conventional electricity, making it ideal for installation in remote locations. Integration with a mobile application also enables fast and efficient responses to incidents in the field.

c. *Channels*

Device distribution through partnerships with government institutions/NGOs, an official website or application as an information and training channel, and innovation exhibitions, competitions, or environmental forums as promotional media.

d. *Customer Relationships*

Training on device usage and installation, periodic technical support (online or direct), and a subscription system for notifications and data updates.

e. *Revenue Streams*

Sales of F-Guard units (per unit), mobile application access licenses (annual subscription), sales of service packages (installation + training + monitoring), and environmental data as a service (if accessed by researchers or other institutions).

f. *Key Resources*

Technology development team (hardware and software), device components (sound sensors, solar panels, communication modules), cloud

infrastructure and mobile application, local partners for distribution and field support, and trained AI models.

g. *Key Activities*

Designing and producing the device, developing the monitoring system and mobile application, training users and performing system maintenance, and improving AI accuracy and data updates. This includes device production, system development, training, and maintenance.

h. *Key Partnerships*

Government and conservation agencies, technology startups or universities (R&D and system development), communication network providers (Sigfox/LoRa), education partners/NGOs, and funding institutions (ESG funds, corporate CSR).

i. *Cost Structure*

- *Fixed* costs: Deep learning model and mobile application development, cloud/server infrastructure (annual subscription), salaries for the development and technical team, initial training and educational material preparation, and device certification and field trials.
- *Variabel costs*: Production of F-Guard units (sensors, solar panels, casing, GPS, etc.), device distribution and installation to locations, commission for distribution partners or field implementers, field technical support and maintenance per device, and communication network usage (Sigfox/LoRa per device).

CHAPTER FOUR

4.1 Solution Provided

F-Guard is a forest monitoring tool based on sound sensors, deep learning, and solar energy to detect illegal deforestation automatically and in real-time. This system operates autonomously in remote locations, without relying on conventional electricity or internet.

Using an edge-cloud approach, the ESP32 handles only sound feature extraction, while classification is performed by a CNN model on a server or gateway. When illegal activity is detected, the device sends location data and notifications to a mobile application via an energy-efficient network (Sigfox/LoRa), enabling a rapid and efficient response.

This solution presents an adaptive, energy-efficient approach to forest monitoring that supports smart city integration through information mobility and community involvement.

4.2 System Architecture and Components

F-Guard is designed as an intelligent, autonomous, and eco-friendly forest monitoring system. It integrates various hardware and software components to automatically detect illegal logging activities in remote areas. The main components that make up the system's architecture include:

a. Sound Sensor (I2S Microphone): Detects sounds from the forest environment, including logging activity.
b. Microcontroller (ESP32): Serves as the main processing unit, handling sound recording, signal pre-processing, and audio feature extraction (MFCC). The ESP32 sends the extracted data to a server or gateway for further classification.
c. GPS Module (Neo-6M): Captures the device's location coordinates when suspicious sounds are detected.
d. Communication Module (LoRa/Sigfox): Transmits data to a gateway or mobile application with a wide range and minimal power consumption.
e. Vibration Sensor: Detects attempts to tamper with or remove the device, acting as an additional security system.
f. Solar Panel and Li-Ion Battery: The primary power source to ensure the device can operate without external electricity supply.

g. F-Guard Mobile Application: Displays device status, location, and real-time notifications to forestry officers.

The device is assembled in a small (approximately 20 × 8 × 6 cm) weather-resistant casing and is discreetly mounted on tree branches to avoid detection by perpetrators.

CHAPTER FIVE

5.1 Product Interface

a. F-Guard Operational Flowchart

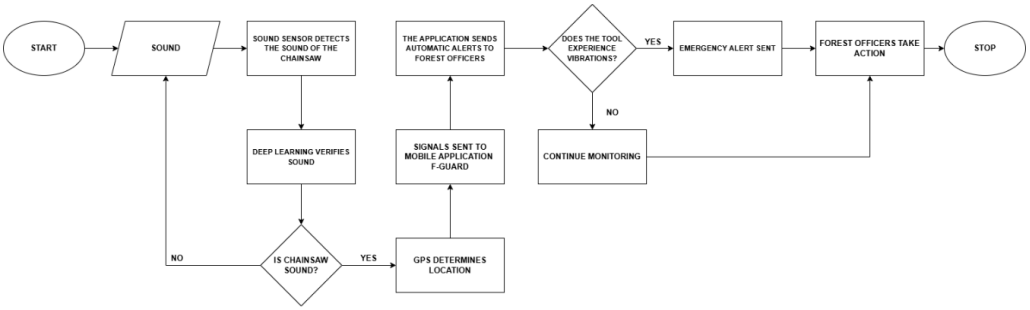


Figure 2. F-Guard Operational Flowchart

The F-Guard system starts when the device receives audio input from its surroundings. A sound sensor detects incoming sounds and identifies whether they resemble chainsaw noises. Once detected, the sound signal is sent to the deep learning module for verification. If the sound does not match the chainsaw sound pattern, the system returns to monitoring mode. However, if the sound is identified as a chainsaw noise, the device's GPS will accurately determine the tool's location.

Once the location is known, the data is sent to the F-Guard mobile application via an energy-efficient communication network. The application then sends an automatic notification to forestry officers as an early warning of illegal activity. Subsequently, the system checks the vibration sensor to determine if the device has experienced shaking or attempted tampering. If the device is detected to have suspicious vibrations, the system will send an additional emergency alert. If there are no vibrations, the system remains active and continues the monitoring process as usual.

Forestry officers receive the information, either through the main notification or emergency alert, and can immediately take action in the field. This process ensures that F-Guard can provide a rapid response to illegal deforestation activities while also protecting the device from potential sabotage.

b. Mock up F-Guard Tool



Figure 3. F-Guard Device Mockup and Its Components

The F-Guard device is a vertical cylinder, approximately 20 × 8 × 6 cm in size, resembling a tree trunk to provide natural camouflage and prevent detection by illegal deforesters. A solar panel is mounted on top of the device as the primary power source, while the outer casing uses weather-resistant (IP65) material. This compact design allows the device to be mounted on tree branches at a height of 2 – 4 meters using clamps or flexible straps.

The main components inside the device include an I2S microphone, ESP32 microcontroller, GPS module, LoRa/Sigfox communication module, Li-Ion battery, and a vibration sensor for device security. All components are arranged vertically for efficient and stable field use.

c. Mock up Mobile App

The F-Guard application is designed to facilitate officers and stakeholders in monitoring the device's condition and receiving real-time information. Here is the interface of the F-Guard application:

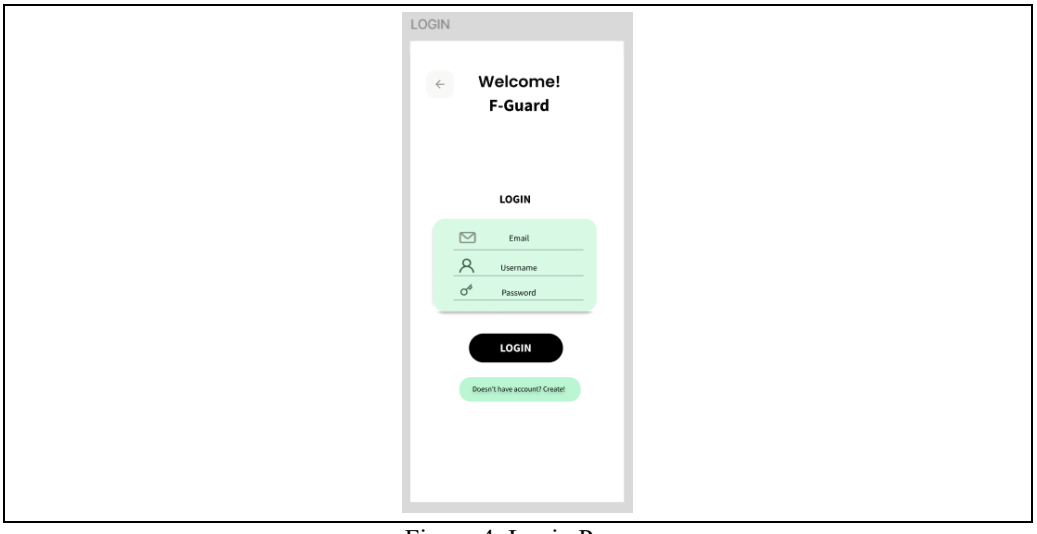


Figure 4. Login Page

Figure 1 illustrates Login Page of the F-Guard application is designed for simplicity and ease of access. With a clean interface, users are welcomed by the

title “Welcome! F-Guard”, followed by a login form requiring Email, Username, and Password. Each input field is supported by intuitive icons for quick identification, all enclosed in a soft green container that adds visual clarity. Once the user fills in their credentials, tapping the “LOGIN” button initiates access to the app’s features. For users without an account, a helpful prompt with a “Create!” button below guides them to the registration page. This page ensures a smooth and welcoming login experience.

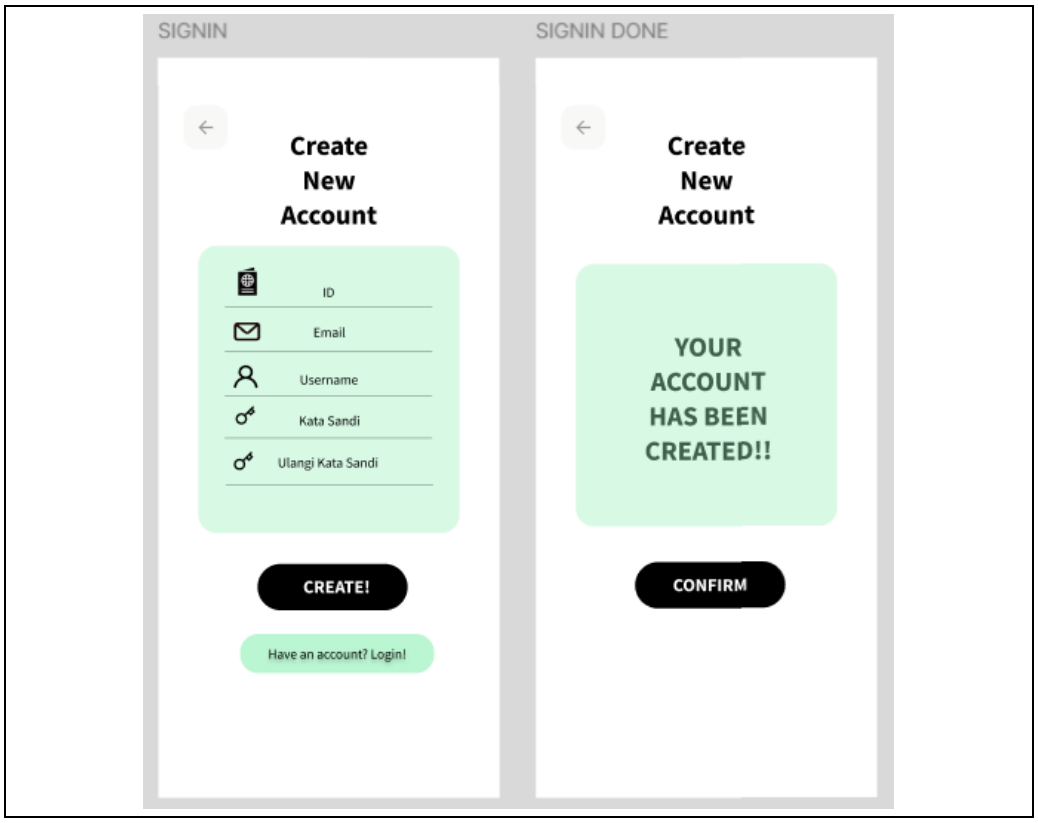


Figure 5. Sign-In Page

Figure 5 illustrates Create New Account screens of the F-Guard application provide a clean and user-friendly interface for onboarding new users. On the left screen, users are prompted to enter essential registration details such as ID, Email, Username, and Password, including a confirmation field to prevent typos. Clear icons alongside each field enhance usability. Once all information is entered, users can tap the prominent “CREATE!” button to register. Upon successful registration, the right screen confirms with a bold message: “YOUR ACCOUNT HAS BEEN CREATED!!”, and a “CONFIRM” button allows users to proceed to the next step seamlessly. This straightforward flow ensures quick and hassle-free account creation.

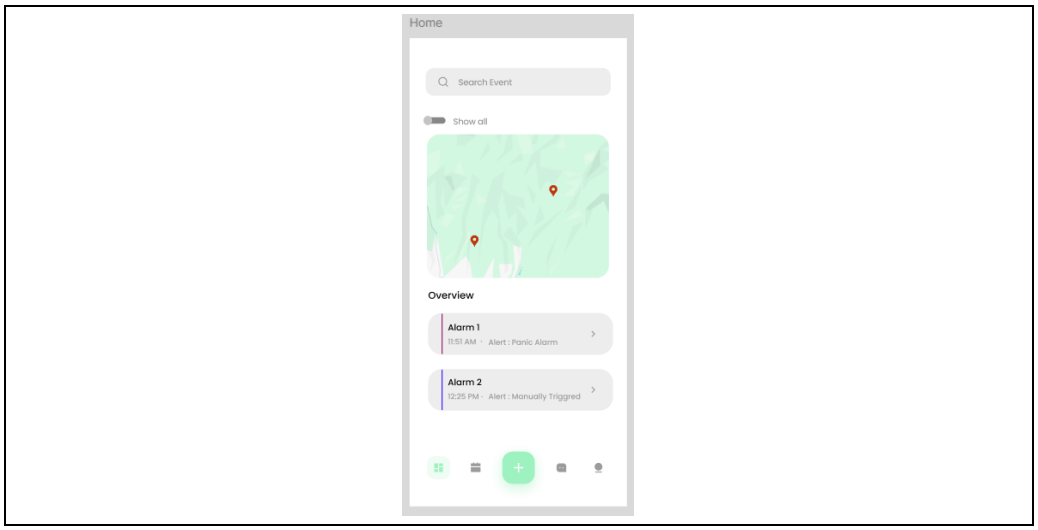


Figure 6. Home Page

Figure 6 illustrates the Home Page of the F-Guard application, which serves as the central dashboard for monitoring forest alert activities. At the top, users can search for specific events and toggle the option to "Show all" alerts. A dynamic map interface displays active alert locations with pinpoint markers, providing an at-a-glance geographic overview. Below the map, an Overview section lists individual alarms, including their time of occurrence, alert type (e.g., Panic Alarm, Manually Triggered), and a quick navigation option for further details. The intuitive layout ensures that users can swiftly detect and respond to potential illegal deforestation events.

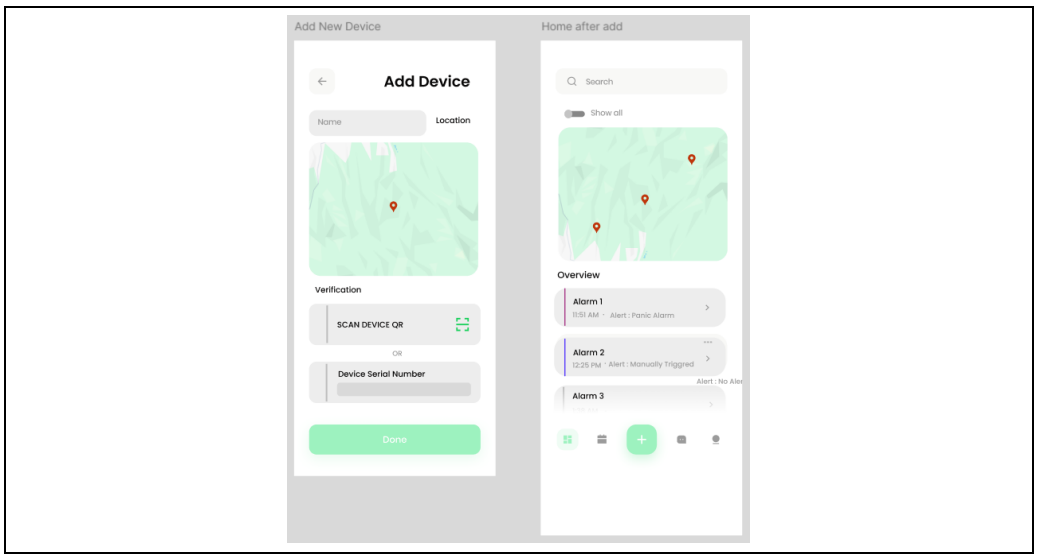


Figure 7. Add Device

Figure 7 illustrates the Add Device and Home After Add screens of the F-Guard application. The left screen enables users to register a new monitoring

device by entering its name, selecting its location on the map, and completing a verification process using either a QR code scan or a device serial number. Once added, the user can confirm the setup by tapping the Done button. The right screen reflects the updated home interface after a device has been successfully added, showing the device's real-time alert activity on a map overview along with a list of alarm events sorted by time and type. This seamless integration ensures quick deployment and monitoring of field sensors.

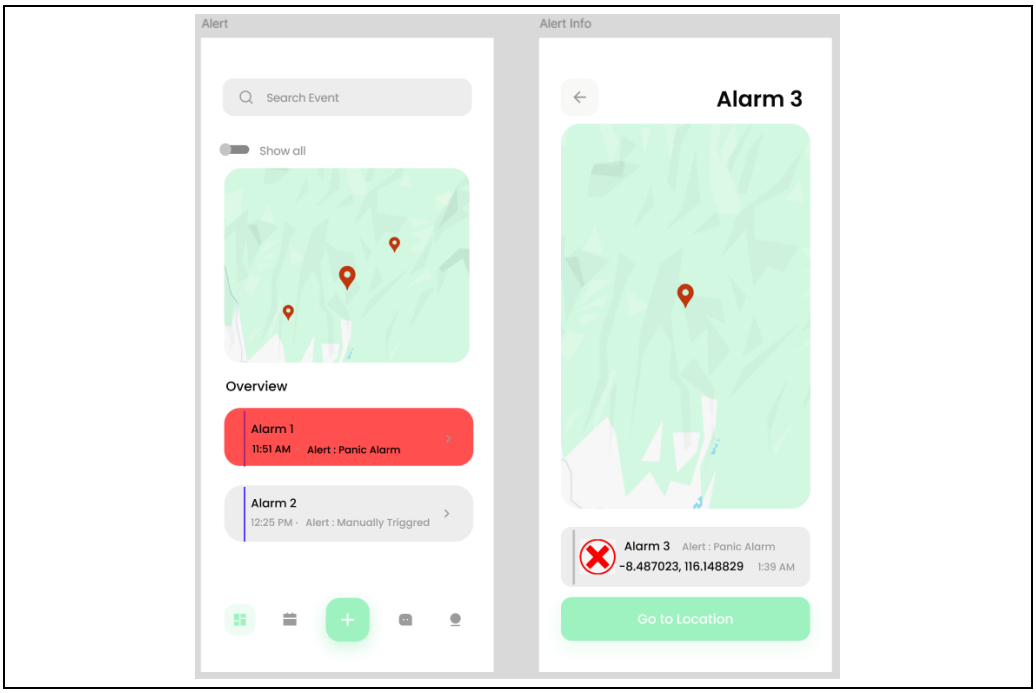


Figure 8. Alert Page

Figure 8 illustrates the Alert and Alert Info Page of the F-Guard mobile application, designed for real-time incident tracking. On the left screen, users can view a map-based overview with markers representing active alerts. Each alert in the Overview list is time-stamped and categorized (e.g., Panic Alarm, Manually Triggered), enabling quick prioritization. The right screen provides detailed information for a selected alert, including GPS coordinates, alert type, and time of occurrence. A "Go to Location" button offers immediate navigation, allowing authorities or responders to take swift action based on precise location data.

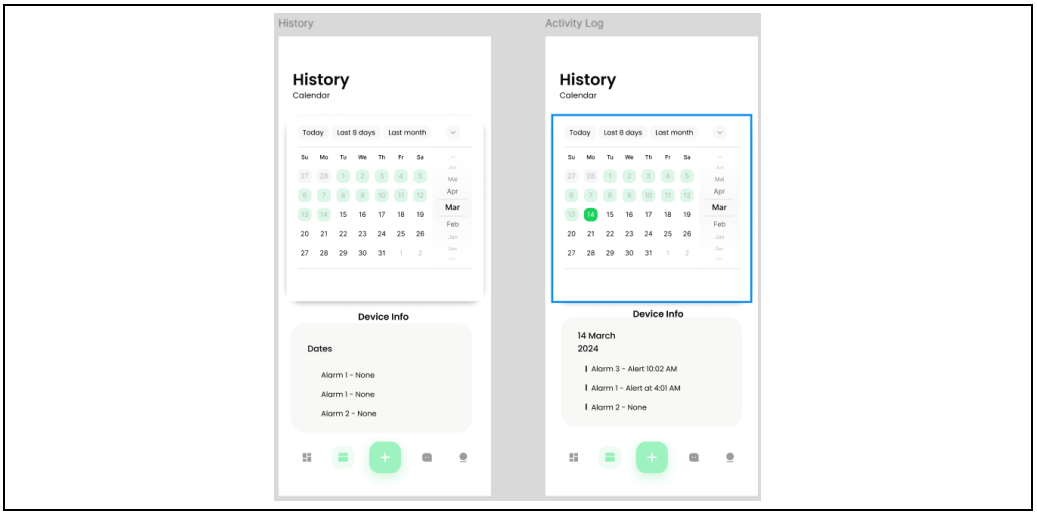


Figure 9. History Page

Figure 9 illustrates the History and Activity Log Page of the F-Guard mobile application. The interface features a calendar-based view that allows users to filter device activity logs by date range—Today, Last 8 Days, or Last Month—to monitor past alerts and events efficiently. The selected dates are visually highlighted in green. Below the calendar, the Device Info section displays detailed logs for a selected date, including specific alarm times and their status. This functionality provides users with clear, time-stamped insights into forest monitoring events, supporting transparency, and traceability.

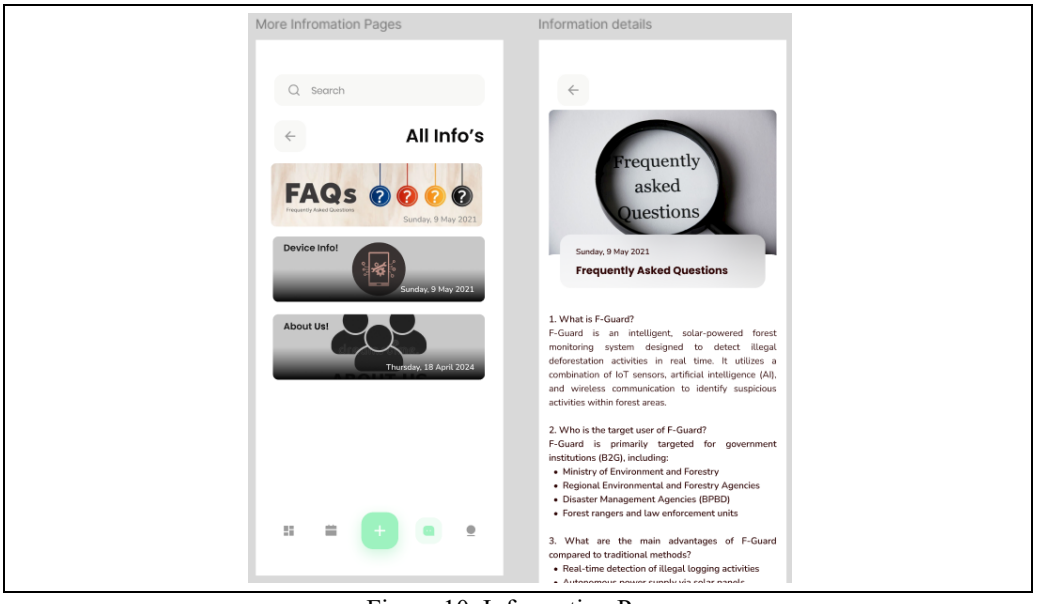


Figure 10. Information Page

Figure 10 illustrates the Information Page and Details View of the F-Guard mobile application. On the left side, the "More Information Pages" screen showcases a scrollable list of categorized information cards, such as FAQs,

Device Info, and About Us, each tagged with a date to indicate the latest update. A search bar at the top allows users to easily find specific topics. On the right side, the "Information Details" screen presents a selected item—in this case, the Frequently Asked Questions (FAQs). It provides detailed insights into the F-Guard system, including its definition, target users (B2G—government institutions), and advantages over traditional methods. The interface is designed for clarity and accessibility, enabling users to easily navigate and understand critical information about the application.

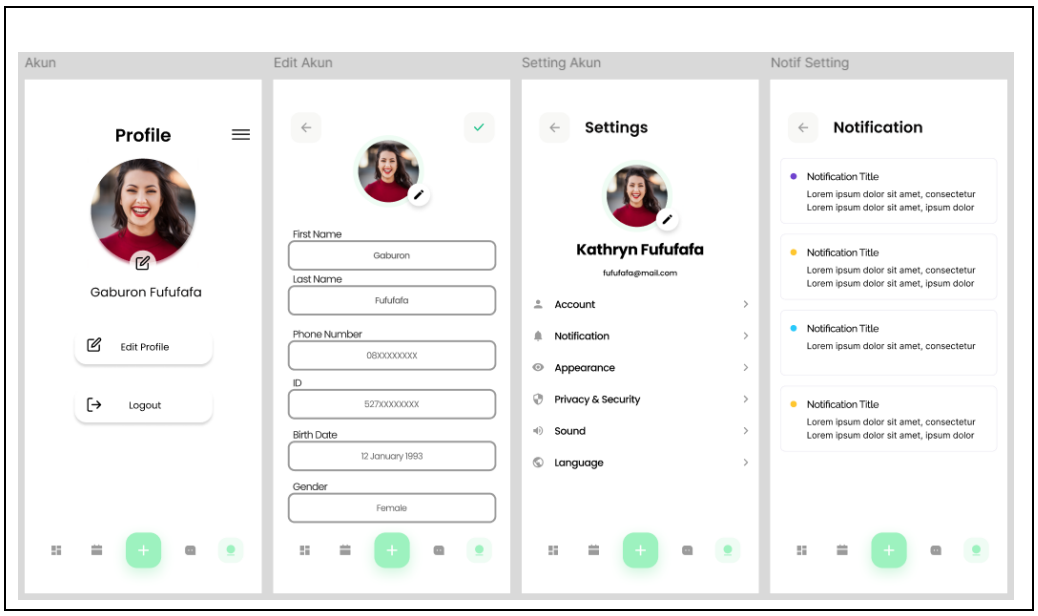


Figure 11. Account and Notification Page

Figure 11 illustrates the Account and Notification Page mockups of the F-Guard mobile application, showcasing a user-friendly interface for profile management and system settings. The first screen displays the user's profile with options to edit details or log out. The second screen enables profile editing, where users can update their personal information such as name, phone number, ID, birth date, and gender. The third screen presents the settings menu, offering access to account, notification, appearance, privacy & security, sound, and language preferences. The fourth screen displays the notification center, listing multiple alerts with color-coded icons and brief descriptions. The design emphasizes clean navigation, minimalistic layout, and accessibility.

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